

SLIDE 1 Customer Shopping Behavior Analytics

Rahul Kumar Sen Data Analyst Portfolio Project

Python | SQL | Power BI

SLIDE 2 Business Problem Statement

A retail company wants to understand customer purchasing behavior and identify factors that influence spending patterns.

The goal is to use transaction data to discover customer preferences, analyze sales performance, and generate insights that support better business decisions.

Business Question:

How can customer shopping data be used to improve customer engagement, optimize product strategy, and increase revenue?

SLIDE 3 Project Overview

Dataset Size: 3,901 Records

Tools Used: • Python • Pandas • SQL • Power BI

Analysis Areas: • Customer Demographics • Product Categories • Purchase Amount • Seasonal Trends • Payment Methods • Customer Behavior

SLIDE 4 Data Cleaning & Preparation

Tasks Performed:

✓ Removed Missing Values

✓ Checked Data Quality

✓ Standardized Dataset

✓ Prepared Data for Analysis

✓ Created Visualizations

SLIDE 5 Python Analysis

Visualizations Created:

- Sales by Category
 - Gender Spending Distribution
 - Seasonal Sales Trends
 - Age Distribution
 - Payment Method Usage
 - Top Purchased Products
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SLIDE 6 SQL Business Analysis

Questions Answered:

- Which customer segment generates the highest revenue?
 - Which categories perform best?
 - What payment methods are most popular?
 - Which products are purchased most frequently?
 - How do demographics affect spending behavior?
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SLIDE 7 Power BI Dashboard

Dashboard Features:

- Total Customers
- Revenue Analysis
- Category Performance
- Customer Demographics

- Subscription Insights

- Interactive Filters

(Add Dashboard Screenshot Here)

SLIDE 8 Key Insights

- ✓ Certain categories contribute significantly to revenue.
 - ✓ Customer spending varies across demographic groups.
 - ✓ Seasonal trends impact purchasing behavior.
 - ✓ Payment preferences reveal customer habits.
 - ✓ Product popularity differs across categories.
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SLIDE 9 Business Recommendations

- Focus marketing on high-performing categories.
 - Improve customer retention strategies.
 - Use seasonal campaigns to increase sales.
 - Optimize inventory based on product demand.
 - Leverage customer behavior insights for decision making.
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SLIDE 10 Skills Demonstrated

- ✓ Python
- ✓ Pandas
- ✓ SQL
- ✓ Power BI
- ✓ Data Cleaning

✓ EDA

✓ Dashboarding

✓ Data Storytelling

SLIDE 11 Thank You

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